COMP 262 - Natural Language Processing and Recommender Systems

Sentiment Analysis of Amazon All Beauty Reviews

Phase 1: Data Exploration, Pre-processing and Lexicon-based Models (VADER and TextBlob)

Team 2

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1. Introduction

This report covers Phase 1 of the COMP 262 project: building a sentiment analyser for products from customers' written reviews using a **lexicon-based approach**. Team 2 was assigned the **Amazon All Beauty** dataset. Phase 1 covers loading and exploring the data, cleaning and labelling it, choosing two sentiment lexicons, pre-processing the text for each lexicon, and building and validating the two lexicon models on a random sample of 1,000 reviews. Phase 2 will add machine-learning models, a recommender-system study and LLM tasks, and will be appended to this report.

All results in this report come from one documented script, **team2_phase1_lexicon_sentiment.py**, which downloads the data, writes every figure into ./figures and every table into ./results, and runs from start to finish in under a minute. A fixed random seed (42) makes every number reproducible.

1.1 Dataset and publishers

We use the **All Beauty 5-core** file (All_Beauty_5.json.gz) from the **Amazon Review Data (2018)** collection published by **Jianmo Ni, Jiacheng Li and Julian McAuley** (UC San Diego), described in their EMNLP 2019 paper (see References). A 5-core subset keeps only users and products with at least five reviews each. As required for Phase 1, we use this small k-core subset; Phase 2 will use the full All Beauty file.

2. Data Exploration

2.1 Attributes and missing values

The raw file has **5,269 reviews and 12 columns**. Table 1 describes every attribute, its type, how many values are missing and what it means. The text fields are almost complete (5 empty reviews), while vote (89.5% missing) and image (98.1% missing) are sparse by nature: a missing vote means nobody marked the review helpful, and a missing image means none was posted.

Table 1. *Data dictionary of the raw All Beauty 5-core file*

Column	Type	Missing	Missing %	Unique	Meaning
overall	int64	0	0.0	5	Star rating given by the reviewer (1-5)
verified	bool	0	0.0	2	True if Amazon verified the purchase
reviewTime	str	0	0.0	883	Review date as text (MM DD, YYYY)
reviewerID	str	0	0.0	991	Anonymised reviewer id
asin	str	0	0.0	85	Amazon product id (one id per size/colour variant)
style	object	645	12.24	408	Dict of variant attributes (Size, Color, ...)
reviewerName	str	0	0.0	905	Display name of the reviewer
reviewText	str	5	0.09	1363	Body of the review
summary	str	5	0.09	1011	Review headline / title
unixReviewTime	int64	0	0.0	883	Review date as a Unix timestamp
vote	str	4717	89.52	24	Number of 'helpful' votes (missing = 0)
image	object	5171	98.14	25	URLs of images posted with the review

2.2 Counts and averages

Table 2. *Headline statistics (raw data)*

Measure	Value
Reviews (rows)	5,269
Unique products (asin)	85
Unique reviewers	991
Average rating	4.77 (median 5, std 0.74)
5-star share	88.2% (4,649 reviews)
Verified purchases	81.2%
Reviews with at least one helpful vote	10.5%
Reviews with images	1.9%
Date range	21 Sep 2004 to 23 Aug 2018

The data is **heavily skewed towards positive ratings**: 88% of rows are 5 stars and only 3.4% are 1 or 2 stars (Figure 1). This imbalance matters for every later step, because a model that always says "Positive" already looks accurate.

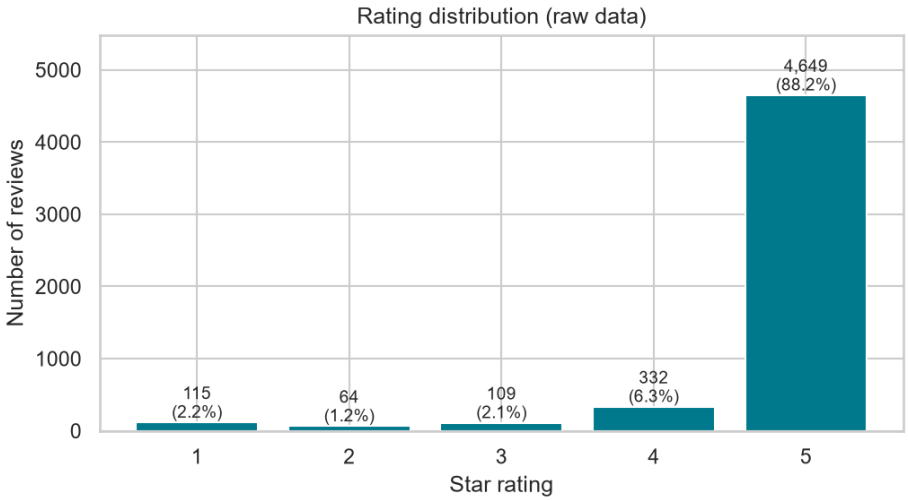


Figure 1. *Rating distribution in the raw data*

2.3 Duplicates: the most important finding

Checking for duplicates changed our view of the whole dataset. Amazon shows the same review on every **size or colour variant** of a product, and each variant has its own asin. So one review written once by one person appears as several rows. We define a duplicate as the same **reviewer + review text + timestamp**.

- **3,890 of 5,269 rows (73.8%) are copies.** Only **1,379 reviews are unique**.
- 720 unique reviews are repeated under 4 different asins, and 135 under 5 (Figure 2).
- No copy ever has a different rating from its original, so the copies are true duplicates and safe to drop.
- 1,044 rows are identical in every field except style and image.

Keeping the copies would make a single review count up to five times in training and testing, and would let the same text appear in both the development and test samples. **All modelling therefore uses the 1,379 unique reviews.** The exploration below shows both views where the difference matters.

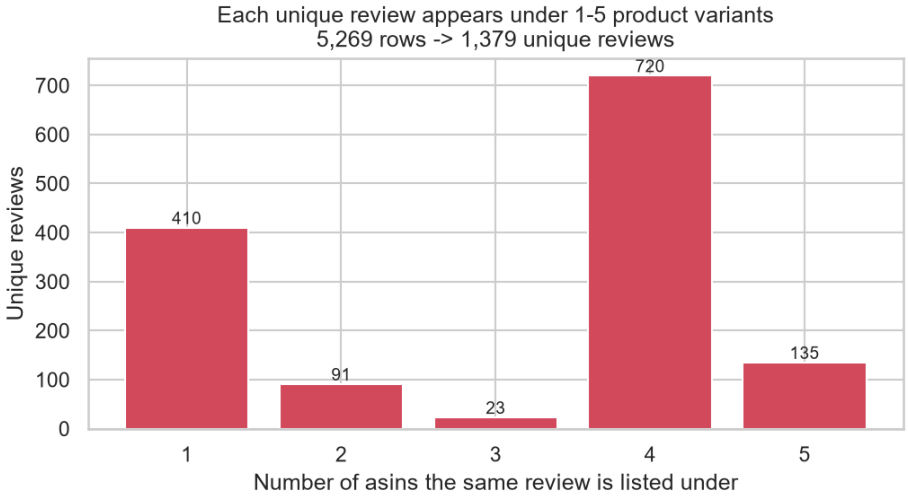


Figure 2. *Number of product variants each unique review is copied to*

2.4 Distribution of reviews across products and per product

Reviews are spread very unevenly across products (Figure 3). In the raw file one product (B0012Y0ZG2) has 1,794 rows, but after de-duplication the largest product has 480 unique reviews. After de-duplication only **63 products** keep any reviews, with a **median of 4** and a mean of 21.9 reviews per product. The **top 2 products hold 62.5%** of all unique reviews and **7 of 63 products hold 80%**, a classic long-tail (Pareto) pattern. Any product-level conclusion is therefore driven by a handful of items.

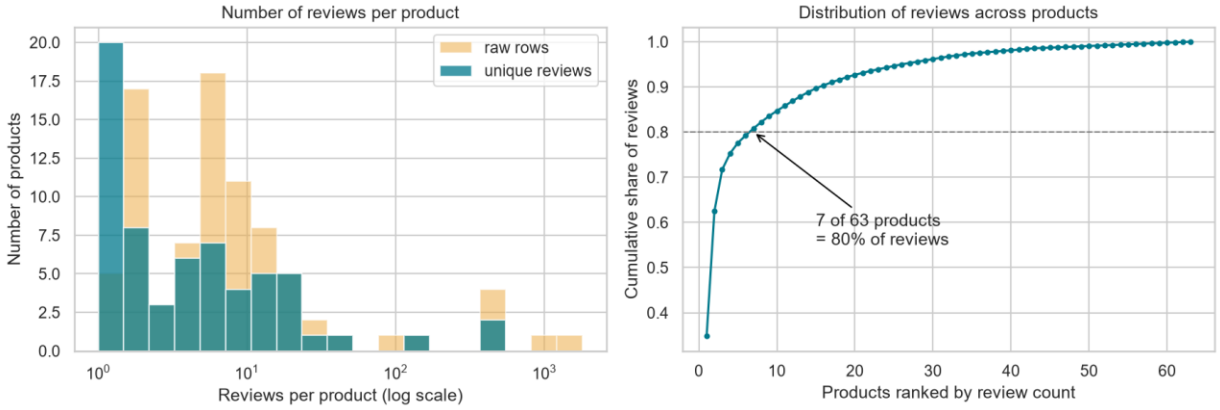


Figure 3. *Left: reviews per product, raw vs unique (log scale). Right: cumulative share of reviews across products*

Figure 4 shows each product's average rating against its number of reviews. Products with few reviews spread from 1 to 5 stars, while popular products all sit between 4.5 and 5. Averages based on a few reviews are unreliable, which is useful to know for the Phase 2 recommender.

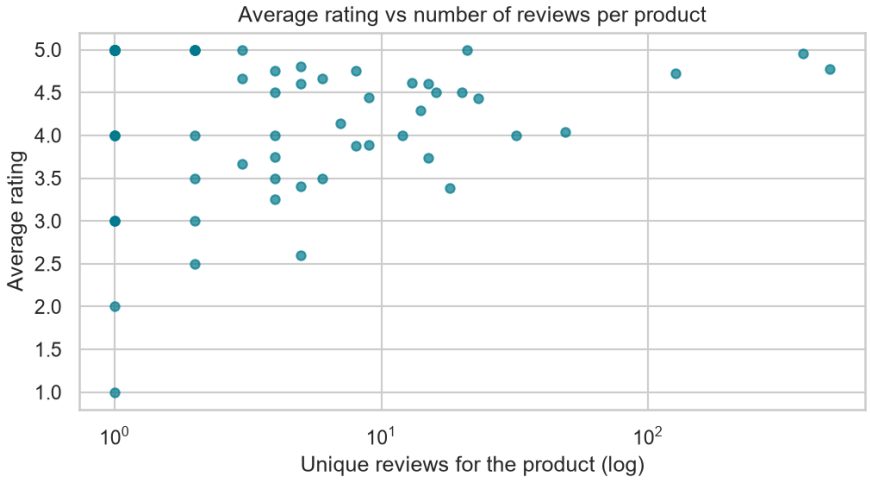


Figure 4. *Average rating vs number of unique reviews per product*

2.5 Distribution of reviews per user

In the raw rows every reviewer has 4 to 12 reviews, as the 5-core filter promises (the minimum is 4 rather than 5 because of small inconsistencies in the published file). After de-duplication the picture is very different: **86.2% of reviewers wrote only one unique review**, the mean is 1.39 and the maximum is 11 (Figure 5). Most of the 5-core guarantee comes from variant copies, so the data is far sparser per user than the name 5-core suggests.

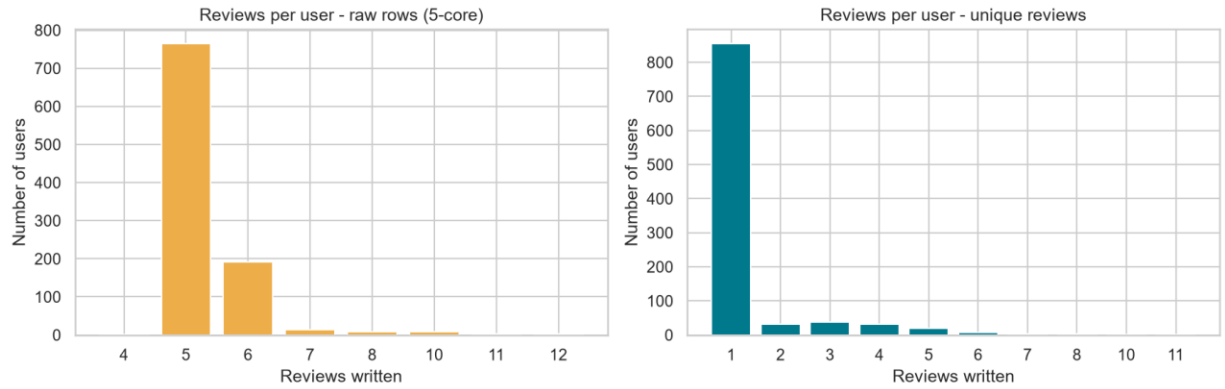


Figure 5. *Reviews per user: raw rows (left) and unique reviews (right)*

2.6 Review lengths and outliers

Unique reviews have a **median of 29 words** (mean 56.8, max 962). The distribution is right-skewed (Figure 6). With the IQR rule (Q1 = 11, Q3 = 71, upper fence = Q3 + 1.5 x IQR = **161 words**), **106 reviews (7.7%) are long outliers**. At the other end, 124 reviews have 3 words or fewer (e.g. "Love it!") and 1 review is empty.

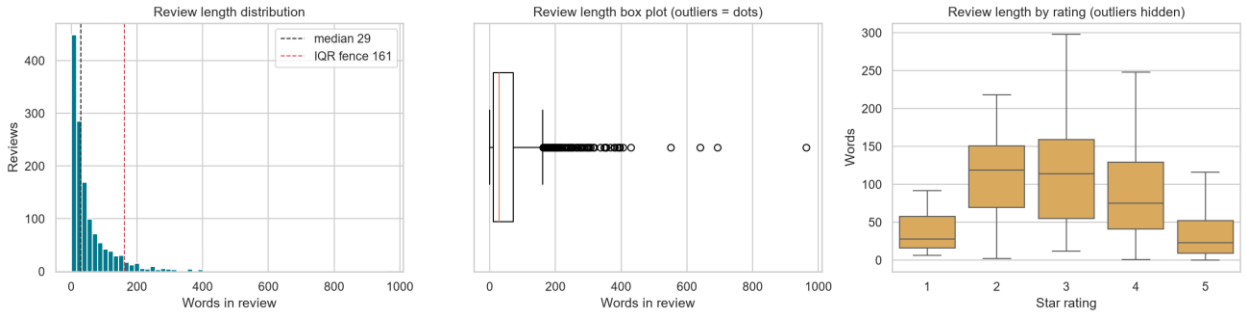


Figure 6. *Review length histogram, box plot and length by star rating*

Table 3. *Review length by rating (unique reviews)*

Rating	Reviews	Mean words	Median words	Max words
1	30	42.4	28.0	200
2	26	114.7	119.0	315
3	64	117.4	114.0	336
4	153	92.9	75.0	405
5	1106	47.4	23.0	962

Length tells us something about sentiment. 2- and 3-star reviews are the longest (median about 115 words), while 5-star reviews are the shortest (median 23) and 1-star reviews are also short (median 28). Mixed opinions take more words to explain ("I liked the smell, but..."), while very happy and very angry customers are brief. This already suggests that **neutral reviews will be the hardest for a lexicon**, because they contain both positive and negative words.

2.7 Other attributes and relationships

- **Verified vs rating:** 90.5% of verified reviews are 5-star (average 4.78), but only 59.9% of unverified ones are (average 4.39). Unverified reviewers give far more 3- and 4-star ratings, which fits free-sample or early-reviewer programmes, where people write more balanced reviews.
- **Helpful votes vs rating:** 1-star reviews are the most likely to get a helpful vote (33.3%), compared with 4.6% of 4-star reviews. Shoppers find negative reviews more informative.
- **Over time:** reviews grow from 2012 and peak in 2017 (about 300 unique reviews). The average rating stays around 4.6 to 4.9 but drops to about 4.0 in 2018 (Figure 7).
- **Style (variant) attributes:** Size appears in 4,358 rows and Color in 378. This confirms that most products are sold in several sizes, which explains the duplicate copies.
- **Summaries:** the review title has a median of 3 words. **307 unique reviews (22.3%) have an automatically generated title** such as "Five Stars" or "One Star". These titles simply repeat the rating (see Section 3.2).

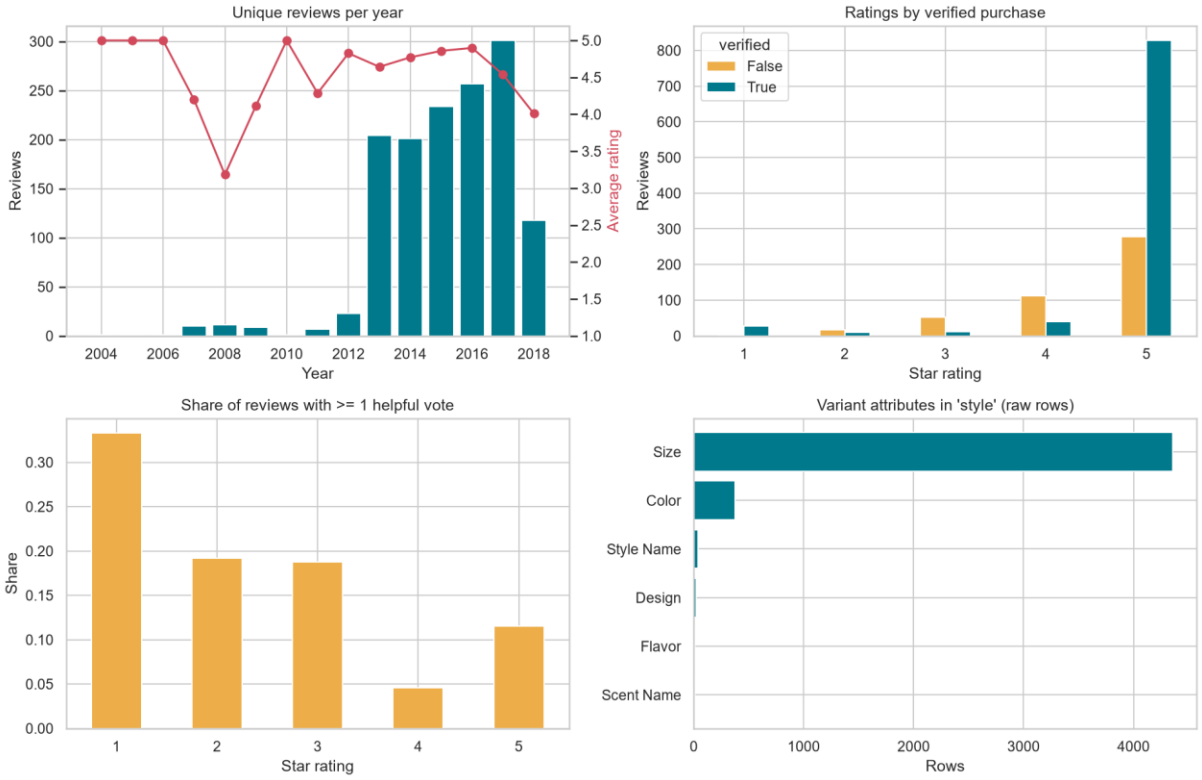


Figure 7. *Reviews per year with average rating, verified status, helpful votes and variant attributes*

2.8 Vocabulary by sentiment

Figures 8 and 9 show the most frequent words (stop words removed) for each class, using the labels defined in Section 3. Positive reviews contain clear sentiment words ("love", "great", "good") next to product words ("hair", "soap", "skin", "scent"). The top 15 negative words contain **no explicitly negative word at all**: "like", "really", "hair", "smell", "water" and "shampoo" lead, because complaints are usually descriptive ("it felt like water"). The neutral list is dominated by "alcohol", "crest" and "mouth", because many 3-star reviews are about one mouthwash product. **"like" is in the top 3 of every class.** VADER treats it as a positive word, so "It smells like chemicals." gets a compound score of +0.36, while in

reviews "like" is often just a comparison. TextBlob scores the same sentence 0. This is an early warning that VADER in particular will over-predict Positive.

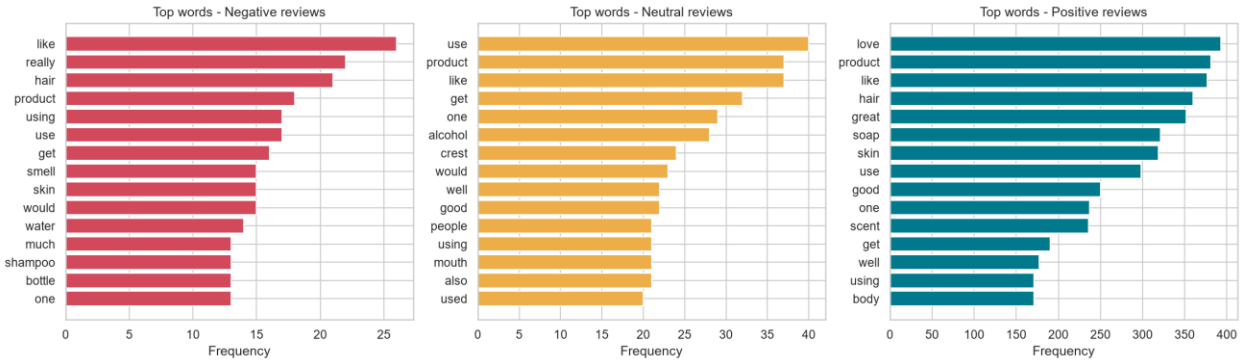


Figure 8. *Top 15 words per sentiment class*

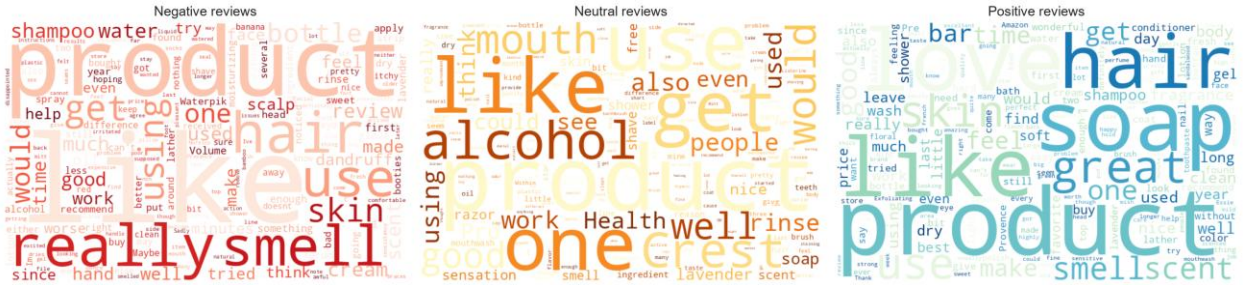


Figure 9. *Word clouds per sentiment class*

2.9 Exploration conclusions

1. Three quarters of the raw rows are duplicates caused by product variants; only 1,379 reviews are unique.
2. The data is extremely imbalanced (91% positive after labelling), so **accuracy is a misleading metric**. We use macro-averaged F1 as the main metric and compare against an always-Positive baseline.
3. Reviews and products follow a long tail: a few products and many one-time reviewers.
4. Neutral and 2-star reviews are long and mixed, which predicts that lexicon models will struggle on the Neutral class.
5. The auto-generated "X Stars" titles leak the label and must not be used as model input.

3. Text Basic Pre-processing

3.1 Labelling

Each review is labelled from its star rating (overall): **4-5 = Positive, 3 = Neutral, 1-2 = Negative**. On the 1,379 unique reviews this gives:

Table 4. *Sentiment labels on unique reviews*

Label	Ratings	Reviews	Share
Positive	4, 5	1,259	91.3%
Neutral	3	64	4.6%
Negative	1, 2	56	4.1%

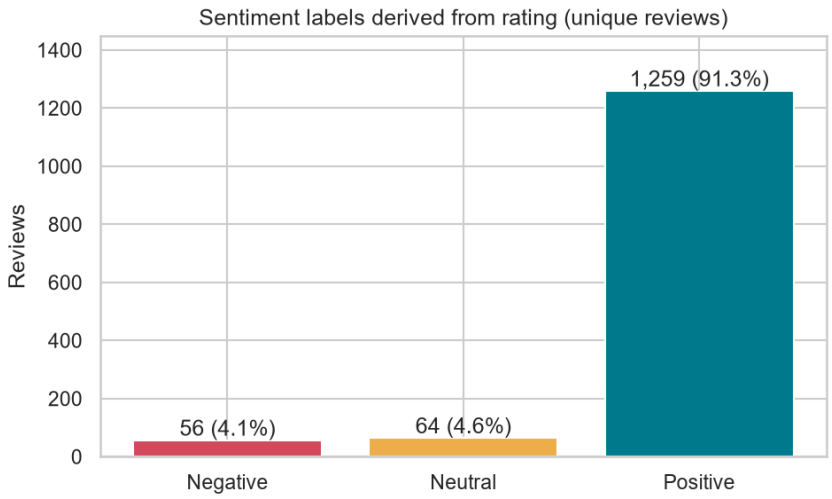


Figure 10. *Sentiment label distribution*

3.2 Column selection

A lexicon model reads only text, so the choice is mostly about which text to score and which columns support cleaning and labelling.

Table 5. *Columns chosen for the sentiment analyser and why*

Column	Used for	Reason
reviewText	Model input	The main opinion text. Holds almost all of the sentiment words.
summary	Model input (prefixed to the body)	Short, opinion-dense headlines such as "Smells heavenly" or "Waste of money" add strong signals. Joined with a full stop so VADER treats it as its own sentence.
summary = "Five Stars" etc.	Removed	Amazon generates these titles from the rating, so they leak the label. They also carry no lexicon sentiment, so keeping them would only add noise.
overall	Label only	Ground truth for the Positive / Neutral / Negative label. Never shown to the models.
reviewerID, unixReviewTime	De-duplication key	Together with reviewText they identify one written review.

Column	Used for	Reason
asin	Exploration only	Product id. Needed in Phase 2 for the recommender.
verified, vote, image, style	Exploration only	Metadata with no opinion words. Useful for exploration but a lexicon cannot use them.
reviewerName, reviewTime	Not used	Personal / redundant (reviewTime duplicates unixReviewTime).

3.3 Outliers

Table 6. *Outlier checks and decisions*

Check	Finding	Decision
Duplicate copies	3,890 rows (73.8%)	Removed - keep one copy of each review
Invalid ratings	0 (all ratings are 1-5)	Nothing to do
Empty text	1 review	Removed - nothing to score
Very short reviews (3 words or fewer)	124	Kept - short but clearly opinionated ("Love it!")
Long reviews (> 161 words, IQR fence)	106 (7.7%)	Kept - they are real reviews and mostly 2-4 star, i.e. the minority classes we need. Their accuracy is checked separately in Section 7.
HTML entities / tags, URLs	10 / 3 reviews	Cleaned (unescaped and stripped) in pre-processing
Non-English text (> 20% non-ASCII)	0	Nothing to do

After these steps the **clean modelling pool has 1,378 unique reviews**.

4. Lexicon Study and Selection

We studied three lexicon packages: **VADER**, **TextBlob** and **SentiWordNet**. Besides reading their documentation, we scored ten probe sentences written in the style of beauty reviews (Table 8) to see how each lexicon handles negation, emphasis, emoticons and contrast.

Table 7. *Comparison of the three lexicons*

Property	VADER	TextBlob	SentiWordNet
Lexicon size	~7,500 entries incl. slang, emoticons, emoji	~2,900 adjectives (pattern)	~117,000 synsets (WordNet)
Built for	Social media / short informal text	General text	General text, per word sense
Negation	Yes (3-word window)	Yes ('not' flips)	No
Intensifiers / CAPS / !!!	Yes / Yes / Yes	Yes / No / No	No / No / No
Emoticons, emoji	Yes	Some emoticons	No
'but' contrast	Yes (weights 2nd clause)	No	No
Needs POS tagging / WSD	No	No	Yes / Yes (hard)
Output	pos/neu/neg + compound [-1,1]	polarity [-1,1] + subjectivity [0,1]	pos/neg/obj per synset
Chosen	YES	YES	No

Table 8. *Probe sentences: VADER compound, TextBlob polarity and SentiWordNet score (-1 to +1)*

Sentence	VADER	TextBlob	SentiWordNet
This moisturizer is good.	0.44	0.7	0.438
This moisturizer is not good.	-0.341	-0.35	0.083
This moisturizer is GOOD!!!	0.671	1.0	0.438
This moisturizer is very good.	0.493	0.91	0.292
The scent is nice, but it broke me out terribly.	-0.827	-0.2	0.271
Love it :)	0.802	0.5	0.5
Meh. It's ok I guess.	0.226	0.5	0.25
Smells amazing, my skin feels so soft.	0.586	0.35	-0.031
Waste of money, it burned my scalp.	-0.421	-0.2	0.0
Not bad at all.	0.431	0.35	-0.625

4.1 What the probes show

- **Negation:** "not good" flips to negative in VADER (-0.34) and TextBlob (-0.35). SentiWordNet still scores it as positive (+0.08) because it scores words on their own.
- **Emphasis:** "GOOD!!!" is stronger than "good" in VADER (0.67 vs 0.44), because VADER has rules for capital letters and exclamation marks. SentiWordNet ignores both.

- **Contrast:** "The scent is nice, but it broke me out terribly" is clearly negative. VADER gets this right (-0.83) because it gives more weight to the clause after "but". TextBlob is weakly negative (-0.20) and SentiWordNet says positive (+0.27).
- **Emoticons:** "Love it :)" gets a high score from VADER (0.80).
- **SentiWordNet fails in obvious ways:** "Not bad at all" is -0.63 and "Smells amazing, my skin feels so soft" is -0.03, because it picks the most common sense of each word without any context.

4.2 Choice: VADER and TextBlob

VADER was chosen because it was built for short, informal, opinion-heavy text like product reviews. It handles negation, intensifiers ("very", "so"), capital letters, exclamation marks, emoticons and "but" contrasts, and it needs no training data. Its compound score from -1 to +1 has documented thresholds.

TextBlob was chosen as the second model because it works differently. Its lexicon (from the Pattern library) is mostly adjectives such as "soft", "greasy" and "amazing", which suit product reviews, and it handles negation and intensifiers. It also gives a **subjectivity** score that we can use later. Because the two models make different mistakes, comparing them tells us more than comparing two similar models would.

SentiWordNet was not chosen. It needs POS tagging and word-sense disambiguation, it has no negation or emphasis rules, and it failed the simplest probes.

5. Pre-processing for Each Model

Lexicon models do not learn, so pre-processing has to keep exactly the signals each lexicon's rules use. The "classic" NLP pipeline (lower-case, remove punctuation and stop words, lemmatise) can **destroy** those signals: NLTK's stop-word list contains "not", "no", "very", "too" and "but". We therefore designed a different pipeline for each model and tested it (Section 5.3).

5.1 Steps applied to both models

Table 9. *Shared cleaning steps*

Step	Justification
Combine summary + body	Adds the opinion-dense headline (Section 3.2). Auto-generated "X Stars" titles are removed first.
Unescape HTML (& -> &) and strip tags	10 reviews contain HTML such as or <a href...>. These are not words and only confuse tokenisation.
Remove URLs	3 reviews contain links. They carry no sentiment.
Squeeze repeated characters ("sooooo" -> "soo")	Stretched words do not match lexicon entries. Two letters are kept so the emphasis is not lost.
Normalise whitespace	Removes line breaks and double spaces left by the cleaning.

5.2 Model-specific steps

Table 10. *Pre-processing steps per model*

Step	VADER	TextBlob	Why
Lower-case	No	Yes	VADER uses CAPS as emphasis ("GOOD" > "good"). TextBlob's lexicon is lower-case and has no CAPS rule.
Punctuation	Keep	Remove	VADER uses "!" and "?" for intensity and "but" clauses. TextBlob gains nothing from them.
Expand contractions	No	Yes	VADER's lexicon already contains "isn't" and "don't". TextBlob needs "is not" as a separate word for its negation rule.
Stop words	Keep all	Remove, but keep negations and intensifiers	VADER's rules depend on "not", "very" and "but". TextBlob only flips polarity when the negation comes right before the sentiment word, so removing other stop words brings "not ... good" together.
Lemmatisation	No	Yes	VADER's lexicon already has inflected forms. Lemmatising maps "smells" and "scents" onto TextBlob's base entries.
Emoticons / emoji	Keep	Removed with punctuation	VADER scores them. TextBlob mostly ignores them.

Example review: "I didn't feel like it was actually shampoo. So watered down..."

- **VADER input:** unchanged. The text was already clean, and case, punctuation and "didn't" are all kept.
- **TextBlob input:** "not feel like actually shampoo so watered not feel like actually shampoo tried twice threw away"

- **Aggressive pipeline:** "feel like actually shampoo watered feel like ..." - the negation is lost, so the meaning reverses.

5.3 Ablation: testing the choices

To justify these choices with data, we scored the **development set** (the 378 clean reviews **not** in the 1,000-review test sample) with five pipelines, using default thresholds. Using a separate development set means the test sample never influenced any decision.

Table 11. *Pre-processing ablation on the development set (378 reviews)*

Model	Pre-processing	Accuracy	Macro-F1	Chosen
VADER	P0 raw text	0.892	0.455	
VADER	P1 light clean (keep case, punct, stop words)	0.892	0.455	yes
VADER	P2 lower + contractions + no punct	0.892	0.456	
VADER	P3 P2 + drop stop words except negations + lemma	0.881	0.437	
VADER	P4 P2 + drop ALL stop words + lemma	0.892	0.416	
TextBlob	P0 raw text	0.889	0.46	
TextBlob	P1 light clean (keep case, punct, stop words)	0.889	0.46	
TextBlob	P2 lower + contractions + no punct	0.886	0.464	
TextBlob	P3 P2 + drop stop words except negations + lemma	0.894	0.506	yes
TextBlob	P4 P2 + drop ALL stop words + lemma	0.894	0.492	

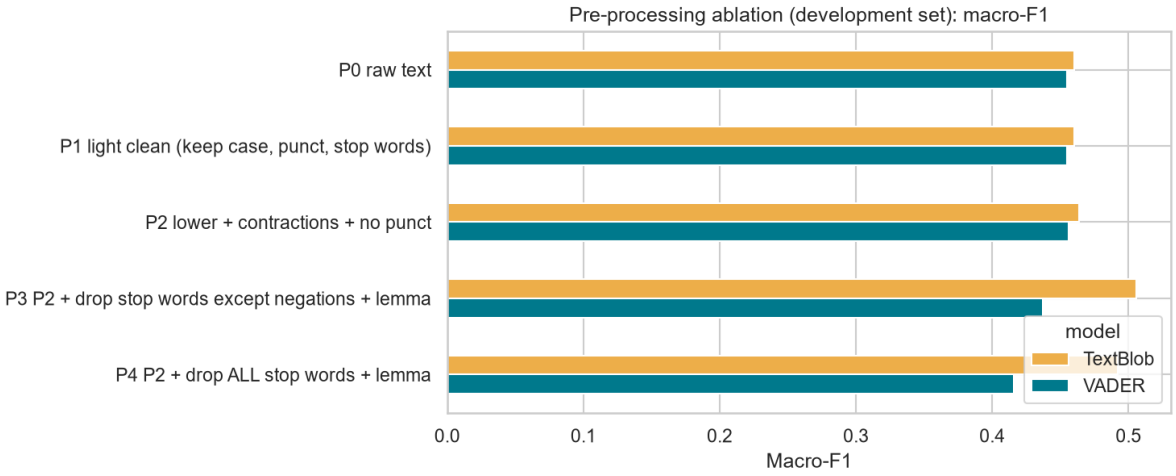


Figure 11. *Macro-F1 of each pre-processing pipeline on the development set*

- **VADER:** light cleaning (P1) is as good as any option, and anything that removes stop words hurts. Removing all stop words drops macro-F1 from 0.455 to 0.416. The dataset is already fairly clean, so P1 gives the same score as raw text, but it protects against the HTML and URLs we found.
- **TextBlob:** P3 (remove stop words but keep negations, then lemmatise) is clearly best (0.506 vs 0.460 for light cleaning). P4, which also removes "not", is worse (0.492). This confirms that **negations must be kept**.

- Same input data, opposite best pipelines: **pre-processing has to match the model**.

6. Sampling and Lexicon Models

6.1 Random sample of 1,000 reviews

We randomly selected **1,000 reviews** from the clean pool of 1,378 unique reviews (pandas sample, random_state = 42). The remaining 378 reviews form the development set used in Sections 5.3 and 6.4. Because we sample from unique reviews, no review can appear in both sets.

Table 12. *Class counts in the test sample and development set*

Set	Negative	Neutral	Positive	Total
Test sample	38 (3.8%)	51 (5.1%)	911 (91.1%)	1,000
Development set	18	13	347	378

The sample keeps the natural class balance, as the assignment asks for a random sample. We did not oversample the minority classes, so the results reflect what a lexicon model would face on real Amazon beauty reviews.

6.2 Model 1: VADER

- **Package:** vaderSentiment 3.3 (Hutto and Gilbert, 2014).
- **How it works:** each word is looked up in a human-rated lexicon of about 7,500 entries scored from -4 to +4. Five rules then adjust the scores: punctuation ("!" adds intensity), capitals, degree words ("extremely", "slightly"), "but" (the second clause gets more weight), and negation within three words ("not" flips and dampens). The adjusted sum is normalised to a **compound score in [-1, +1]**.
- **Decision rule (default):** compound >= 0.05 Positive, <= -0.05 Negative, otherwise Neutral (the thresholds recommended by the authors).
- **Assumptions:** sentiment is carried by individual words and short-range rules, and every sentence of a review counts equally.

6.3 Model 2: TextBlob

- **Package:** TextBlob 0.20 with its default PatternAnalyzer (lexicon from the Pattern library, De Smedt and Daelemans, 2012).
- **How it works:** about 2,900 English words (mostly adjectives) have polarity in [-1, 1] and subjectivity in [0, 1]. TextBlob finds these words, multiplies by a preceding intensifier ("very" x 1.3), multiplies by -0.5 when a negation directly precedes the word, and **averages** the scores over the text. The output is **polarity in [-1, +1]** plus subjectivity.
- **Decision rule (default):** polarity > 0 Positive, < 0 Negative, = 0 Neutral.
- **Assumptions:** opinion is mostly expressed by adjectives, and averaging works for long texts.

6.4 Fine-tuning: threshold search

Lexicon models have no weights to train, but the **thresholds** that turn a score into a class can be tuned. 3-star reviews are usually worded mildly positively, so the default neutral band (almost zero width) is too narrow. We grid-searched the positive and negative thresholds on the **development set** to maximise macro-F1:

- **VADER:** Positive >= 0.40, Negative <= -0.40 (development macro-F1 0.555).
- **TextBlob:** Positive >= 0.075, Negative <= 0.05 (development macro-F1 0.596).

Both tuned and default versions are evaluated on the test sample in Section 7. Figure 12 shows why tuning is needed: the score distributions of the three classes overlap heavily, and most Neutral reviews score above the default positive threshold.

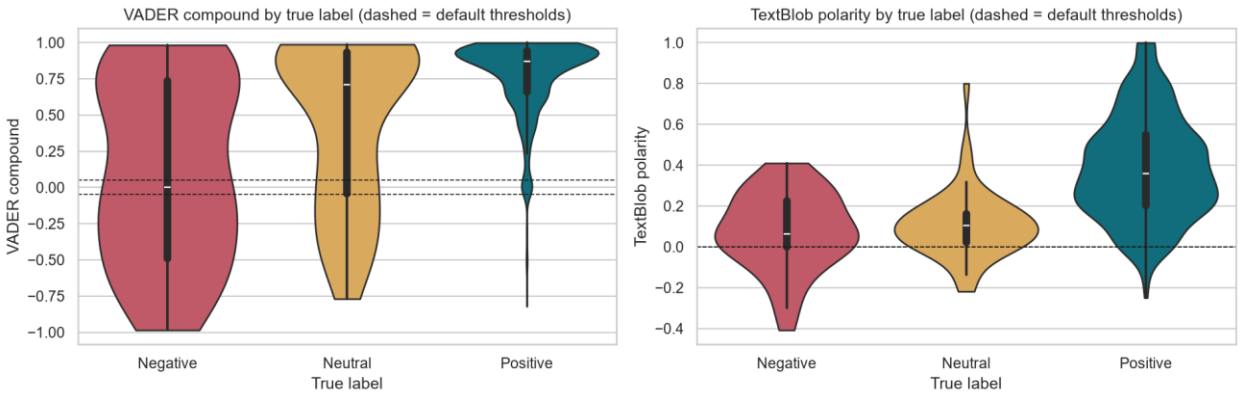


Figure 12. *Score distribution by true label on the test sample (dashed lines = default thresholds)*

7. Validation and Comparison

Both models were run on the 1,000-review test sample and compared with the true labels using accuracy, macro and weighted precision / recall / F1, per-class F1 and confusion matrices. **Macro-F1 is our main metric**: it gives the three classes equal weight, so a model cannot score well by ignoring the 9% of non-positive reviews. A majority-class baseline (always "Positive") is included for reference.

Table 13. *Comparison table on the 1,000-review test sample*

Model	Accuracy	Macro P	Macro R	Macro-F1	Weighted F1	F1 Neg	F1 Neu	F1 Pos
Majority baseline (always Positive)	0.911	0.304	0.333	0.318	0.869	0.000	0.000	0.953
VADER (default thresholds)	0.891	0.471	0.467	0.469	0.884	0.432	0.023	0.951
TextBlob (default thresholds)	0.866	0.403	0.405	0.404	0.861	0.234	0.044	0.933
VADER (tuned (0.4, -0.4))	0.859	0.520	0.487	0.490	0.872	0.361	0.177	0.932
TextBlob (tuned (0.075, 0.05))	0.834	0.385	0.463	0.391	0.848	0.225	0.030	0.919

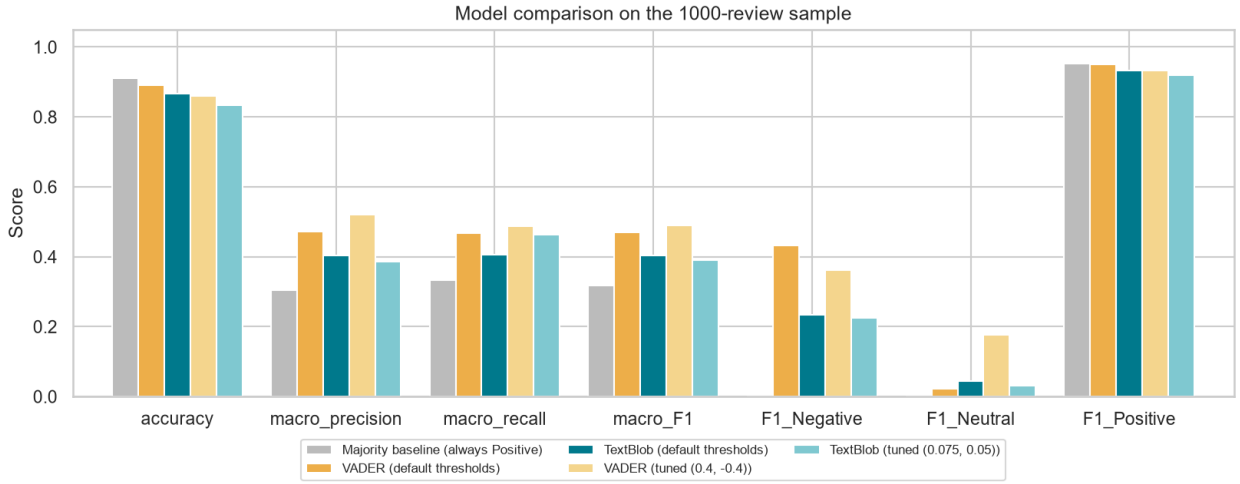


Figure 13. *Main metrics for each model and the baseline*

7.1 Per-class results (default thresholds)

Table 14. *Classification report per class*

Class	Support	VADER P	VADER R	VADER F1	TextBlob P	TextBlob R	TextBlob F1
Negative	38	0.444	0.421	0.432	0.231	0.237	0.234
Neutral	51	0.028	0.020	0.023	0.051	0.039	0.044
Positive	911	0.942	0.959	0.951	0.927	0.939	0.933
Macro avg	1000	0.471	0.467	0.469	0.403	0.405	0.404

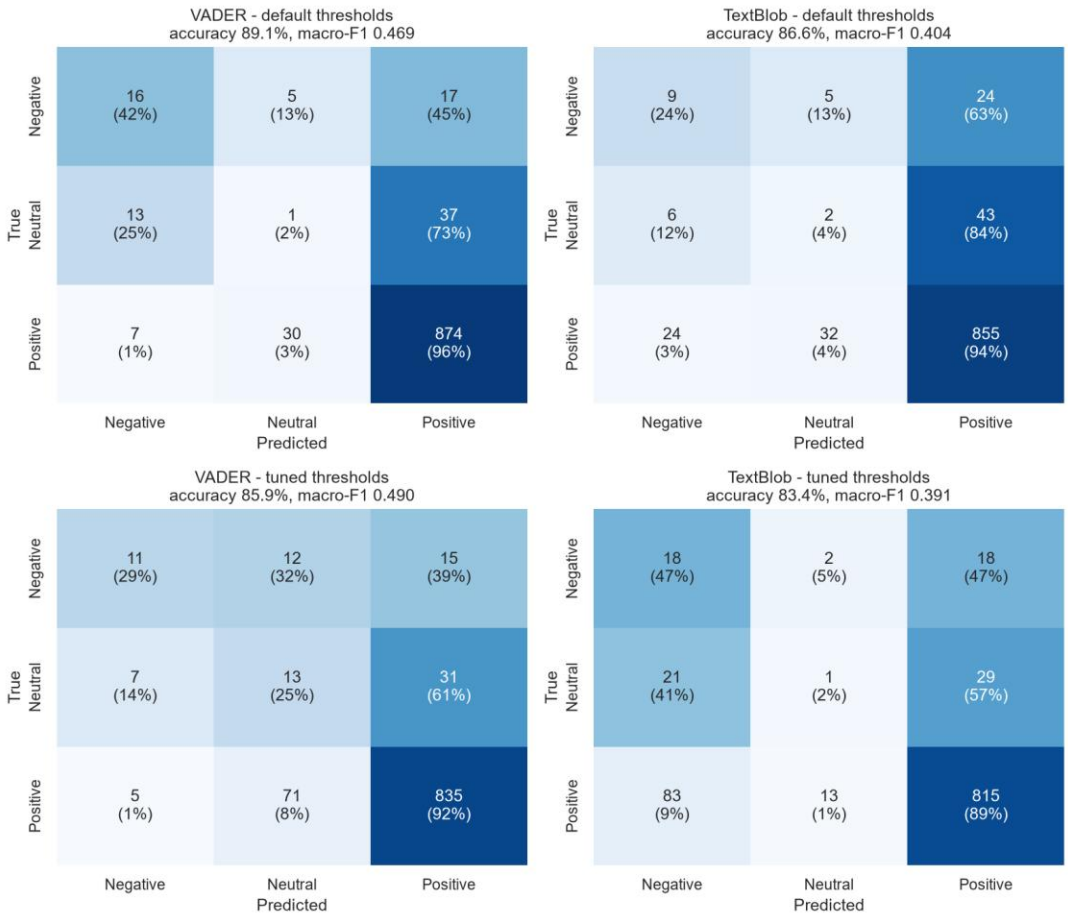


Figure 14. *Confusion matrices (row % = recall per true class)*

7.2 Interpretation

- **VADER is the better model.** With default thresholds it reaches macro-F1 **0.469** vs 0.404 for TextBlob, and it is better on Negative reviews (F1 0.43 vs 0.23). Its rules for "but", emphasis and negation capture more of the meaning of complaints.
- **Accuracy is misleading.** The always-Positive baseline has the highest accuracy (91.1%) but a macro-F1 of only 0.318 and finds no negative or neutral review at all. Both lexicon models are slightly less accurate than the baseline but far more useful.
- **Neutral is almost never detected** (F1 0.02 to 0.04 with default thresholds). 73% of neutral reviews are called Positive by VADER and 84% by TextBlob. 3-star reviews mix praise and complaints, and politely worded criticism still contains many positive words.
- **Threshold tuning helps VADER** (macro-F1 0.469 -> 0.490, Neutral F1 0.02 -> 0.18), at the cost of accuracy (89.1% -> 85.9%) because more positive reviews fall into the wider neutral band. **For TextBlob the tuned thresholds did not carry over** to the test sample (0.404 -> 0.391): the development set has only 13 neutral reviews, which is too few to tune reliably. We therefore report the **default thresholds as the main result** and the tuned ones as an experiment.
- **Errors are lopsided:** the most common mistake for both models is predicting Positive for a Neutral or Negative review, which matches the score overlap in Figure 12.

7.3 Agreement between the models

The two models give the same label on **91.8%** of reviews, but **Cohen's kappa is only 0.42** and their raw scores correlate weakly (**r = 0.27**, Figure 15). Most of the agreement comes from easy positive reviews: both are right on 85.0% of reviews and both are wrong on 9.3%. They disagree mostly on exactly the hard cases, which suggests that combining them, or using a learned model in Phase 2, could help.

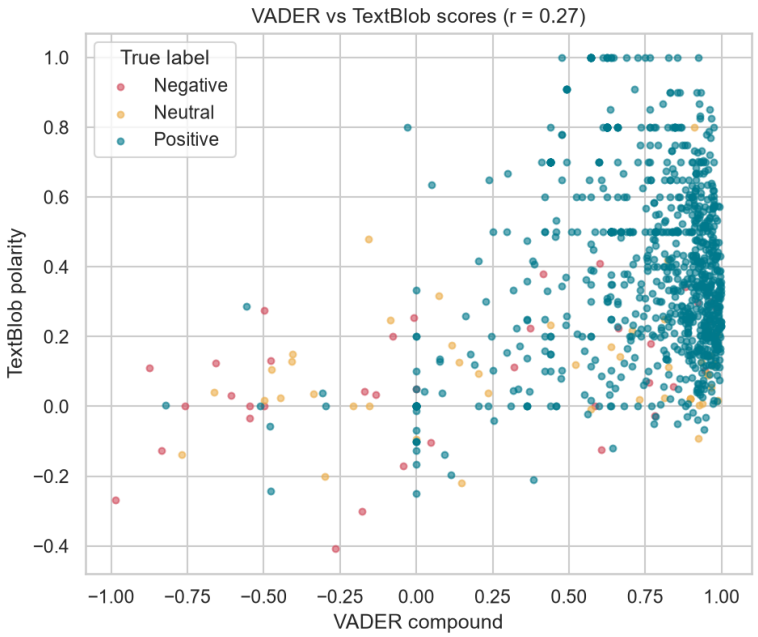


Figure 15. *VADER compound vs TextBlob polarity for each test review, coloured by true label*

7.4 Effect of review length (the long-review outliers)

Table 15. *Accuracy by review length on the test sample*

Words in review	Reviews	VADER accuracy	TextBlob accuracy
1-10	253	0.897	0.866
11-30	268	0.948	0.922
31-70	220	0.891	0.882
71-150	164	0.878	0.829
150+	95	0.737	0.737

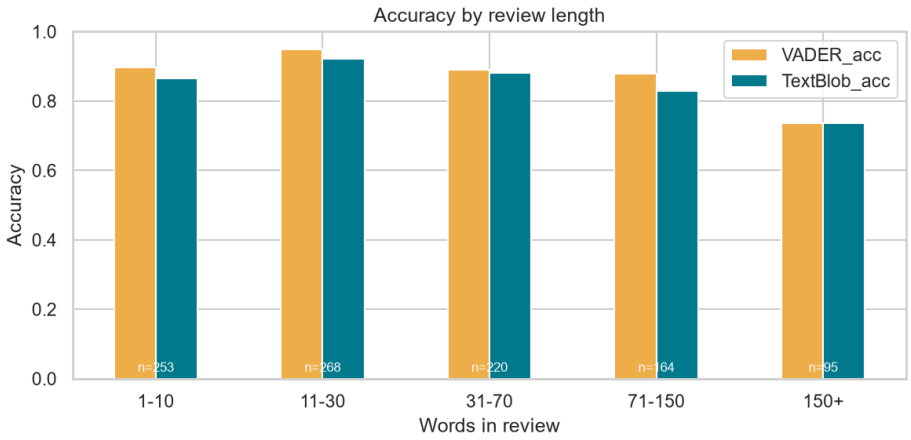


Figure 16. *Accuracy by review length*

Both models work best on reviews of 11-30 words (VADER 94.8%) and get worse on **reviews over 150 words (73.7% for both)**. The long outliers we kept in Section 3.3 are mostly 2-4 star reviews that describe pros and cons in detail. VADER adds up many mildly positive words and saturates near +1, while TextBlob's average gets diluted towards zero. This is a real weakness of lexicon models, not a data error, so keeping these reviews gives an honest estimate.

7.5 Error analysis

The most confident mistakes (Table 16) show recurring patterns:

Table 16. *Examples of confident errors*

Model	True	Predicted	Score	Review (start)
VADER	Neutral	Positive	0.987	Concerned About Active Ingredient. I thought this rinse might be a more palatable alternative to Listerine Total Care. But with average performance a...
VADER	Negative	Positive	0.983	Not So Great for Volume ... Both times that I experimented with the product, I found the results unsatisfactory. It made my hair a bit smoother, but I...
VADER	Neutral	Positive	0.97	YES It Works Quite Well ~ But Unpleasent Odor ~. I just recently started shaving again with an electric razor after 40 years of blades. I bought the.....
TextBlob	Neutral	Positive	0.48	Color doesn't look good on. Way to light. You have to put on 4 coats and it still doesn't look very good. It looks nice in the bottle but it's too pal...
TextBlob	Negative	Positive	0.41	smells great for 5 minutes. I am very dissapointed in this. It smells so good at first but the smell goes completely away after 5 mionutes....
TextBlob	Negative	Positive	0.38	It didn't work for my psoriasis. I really like using organic products and this product had awesome reviews. Ive been seeing a dermatologist for my pso...

- **Polite criticism:** "Not So Great for Volume... It made my hair a bit smoother, but..." contains many positive words, so VADER scores it +0.98 even though it is a 2-star review.
- **Sentiment in the title but not the body:** "I Am Not Happy With the Explanation..." followed by a long, neutral description.
- **Misspellings:** "dissapointed" and "Unpleasent" are not in either lexicon, so the negative word is missed.
- **Domain meaning:** "Didn't smell like redcurrant and basil" is a complaint about a beauty product, but contains no negative word.

7.6 Robustness check on all unique reviews

Because the test sample is a single random draw, we also ran both models on all 1,378 unique reviews. The results are almost the same (Table 17), so the 1,000-review sample is representative.

Table 17. *Default-threshold models on all 1,378 unique reviews*

Model	Accuracy	Macro-F1	F1 Neg	F1 Neu	F1 Pos
VADER (all unique reviews)	0.891	0.465	0.429	0.018	0.949
TextBlob (all unique reviews)	0.874	0.437	0.339	0.034	0.937

8. Conclusions

1. **Data quality first:** 73.8% of the All Beauty 5-core rows are copies of the same review under different product variants. De-duplicating reduces the data to 1,379 reviews and changes the per-user and per-product statistics a lot.
2. **The task is very imbalanced** (91% positive), so we judge models by macro-F1 and against a majority baseline rather than by accuracy.
3. **VADER outperforms TextBlob** on this data: macro-F1 0.469 vs 0.404 with default thresholds, and 0.490 with tuned thresholds. It is especially better on negative reviews, thanks to its contrast, negation and emphasis rules.
4. **Pre-processing must match the model:** light cleaning is best for VADER, while TextBlob improves when stop words are removed but negations are kept. Removing negations hurts both.
5. **Lexicons cannot detect neutral reviews** and struggle with long, mixed reviews. These are the clearest targets for the machine-learning models in Phase 2, which can learn that "but it broke me out" outweighs "smells nice" in this domain.
6. For Phase 2 we will reuse the de-duplicated data and the same 1,000-review sample, so the lexicon and machine-learning models can be compared on exactly the same reviews.

9. Assumptions

- The star rating is the true sentiment of the review. In reality some 3-star reviews are clearly positive or negative in tone, which caps how well any text model can score.
- Reviews with the same reviewer, text and timestamp are the same review, even when they are listed under different asins.
- All reviews are in English (no review had more than 20% non-ASCII characters).
- Auto-generated "X Stars" summaries are written by Amazon, not the reviewer, so they are removed from the model input.
- The random sample (seed 42) is representative of the dataset. This was checked against all unique reviews in Section 7.6.
- The package default thresholds are used as the main result. Tuned thresholds come only from the 378-review development set.

10. References

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Appendix 1: Project Plan

Update weekly. Owners are placeholders for the team to fill in.

Table 18. *Project plan*

Week	Task	Owner	Status
Week 3	Teams formed, dataset assigned (All Beauty)	All	Done
Week 4	Download data, initial exploration	[Name]	Done
Week 5	Checkpoint 1: exploration and pre-processing	[Name]	Done
Week 6	Checkpoint 2: lexicon study and per-model pre-processing	[Name]	Done
Week 7	VADER / TextBlob models, validation, comparison	[Name]	Done
Week 8	Report, PowerPoint, Phase 1 demo (Nov 5)	All	In progress
Week 9-12	Phase 2: ML models, recommender study, LLM tasks	[Name]	Planned
Week 14	Phase 2 presentation and final report	All	Planned

Appendix 2: Meeting Register

Table 19. *Meeting register*

Date / time	Attendees	Subjects discussed	Assignments
[dd-mm-yyyy hh:mm]	[names]	Kick-off, dataset review	[who does what]
[dd-mm-yyyy hh:mm]	[names]	Exploration findings, duplicate issue	[who does what]
[dd-mm-yyyy hh:mm]	[names]	Lexicon choice, pre-processing ablation	[who does what]
[dd-mm-yyyy hh:mm]	[names]	Results review, report and slides split	[who does what]

Appendix 3: Running the Code

Files submitted: **team2_phase1_lexicon_sentiment.py** (documented source code), ./figures (16 charts), ./results (CSV tables and run_log.txt with the full console output).

```
pip install pandas numpy matplotlib seaborn nltk scikit-learn vaderSentiment textblob
wordcloud
python team2_phase1_lexicon_sentiment.py
```

The script downloads the dataset automatically (about 0.6 MB) if ./data/All_Beauty_5.json.gz is missing, and needs Python 3.10 or later. It is organised in sections that follow the assignment steps (Step 1 to Step 7), and every function is commented.